Foundations of Data Analysis: Observations, Variables, and Domain Context

Task 7 — Technical Report

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1. Introduction

Before running any pandas commands or building a model, it helps to slow down and ask what a dataset actually is. A table of numbers on its own says nothing. The same value could be a measurement, an error, or nothing at all, depending on what it's attached to. This report covers the vocabulary needed to read a dataset correctly: the difference between an observation and a variable, how categorical data differs from numerical data, and what raw data looks like before anyone cleans or calibrates it. It also looks at why units and experimental context change what a number means, drawing examples from Biochemistry, Electronics, and Mechanical engineering. The aim is to build habits that catch mistakes early, before they turn into bad analysis or a broken model downstream.

2. Core Concepts

2.1 Observation vs. Variable

Data is often organised in a tabular format called a Data Matrix. In that format, an observation or an observational unit is a row of the table, which represents an object (experimental unit) in the derived data. Each object is measured according to various aspects. Each of these aspects is denoted as a variable which is typically represented as a column in the table.

The statistical role of a variable is to analyse and find patterns and correlations in the given data. Observations are in turn the values assigned to these variables to calculate descriptive statistics like mean, median or sum.

Considering the "loan50" data matrix *[2]* which is a data set for 50 randomly sampled loans offered, the variables are: {loan_amount, interest_rate, term, grade, etc.}. Each row in the data matrix represents an offered loan.

2.2 Categorical vs. Numerical Data

Numerical Data is the data which can be assigned to a numerical variable, values of which make sense to add, subtract or take averages. There are two types of Numerical Data. Discrete Numerical Data which can only take numerical values with jumps, for example a variable denoting number of people in a city cannot take decimal or negative input but can only accept whole numbers. Continuous Numerical Data which can take any numerical value, for example a variable denoting unemployment rate in a city can take any non-negative numerical value.

Let us consider a variable which denotes unemployment rate in a city. It will be classified as a numerical variable which can take numerical data as its input as it makes sense to perform numerical analysis on this data. On the other hand, a variable like telephone numbers also takes numerical data as its input but it won't make sense to add, subtract or take averages of phone numbers so it is not considered as a numerical variable.

Let us consider a variable that takes states of a country as input. That variable can only take fixed responses as input and as the responses themselves are categorized, it is called a

Categorical Variable which Categorical Data as input. The possible values of a categorical variable are called the variable's levels.

Some variables seem to be a hybrid. They are categorical but their levels seem to have a natural ordering to them, for example a variable which describes the mean education level of a country. A variable with these properties is called an ordinal variable, while a regular categorical variable without this type of special ordering is called a nominal variable.

2.3 Raw Data vs. Processed Data

Raw data is whatever we collect straight from the source, for example survey answers, transaction logs, sensor readings, social media posts, before anyone touches it. It's unfiltered, comes in a mix of formats, tends to pile up in huge volumes, and usually has errors or duplicates buried in it.

Processed data is what we get after cleaning up Raw Data. Removing duplicates and errors, standardizing formats, running some actual analysis on it. The result is structured, smaller in volume, and tells us averages, trends, patterns we can act on.

Processing data fixes errors, speeds up analysis, gives us something solid to base our decisions and further analysis on. It also lets us combine data from different sources into one coherent picture instead of dealing separate messy sets of raw data.

2.4 Measurement Units and Their Importance

Measurement units are standardized quantities we use to express physical properties like length, mass, time, temperature, and volume. They matter because they give everyone a common language for measuring things accurately. Without that, we get inconsistency and confusion in science, engineering, industry, and daily life. Standardized units cut out ambiguity and let us calculate and compare things reliably and make it possible for people in different regions and fields to work together. Without standardization there are errors and inconsistencies across data interpretation, experiments, manufacturing, and tech development.

2.5 Risks of Comparing Values Across Different Units

Comparing values expressed in different units without proper conversion can lead to inaccurate conclusions, calculation errors, and poor decision-making. Such inconsistencies may result in incorrect data analysis, flawed scientific experiments, engineering failures, and financial losses. In software applications, mismatched units can produce invalid outputs and system errors. Therefore, all values should be converted to a common unit before comparison to ensure accuracy, consistency, and reliability.

For example, the Mars Climate Orbiter burned down in Mar's atmosphere as one team of scientists used metric units and the other team used imperial units and they worked with values of different unit systems without converting values from one to another. This resulted in a huge scientific and financial loss.

2.6 Basic Summary Statistics

Basic summary statistics are numerical measures used to describe the key characteristics of a dataset.

The **mean** is the average value, calculated by dividing the sum of all observations by the total number of observations.

The **median** is the middle value when the data is arranged in ascending or descending order, making it less sensitive to extreme values.

The **minimum** is the smallest value in the dataset, while the **maximum** is the largest value.

The **range** is the difference between the maximum and minimum values and indicates the spread or variability of the data.

The **mode** is the observation that occurs the most number of times in the data.

Together, these statistics provide a quick overview of the dataset's central tendency and distribution.

2.7 When the Mean Can Be Misleading

The mean can be misleading when a dataset contains outliers or combines values from different groups with varying characteristics. Since the mean is the sum of all observations divided by the total number of observations, some extremely large or extremely small observations make the mean too large or too small from a value that actually represents the majority of observations.

For example, consider the salaries ₹30,000, ₹32,000, ₹31,000, ₹33,000, and ₹5,00,000. Although the mean salary is ₹1,25,200, it does not reflect the typical salary of most employees. In such cases, the median provides a more accurate measure of central tendency.

3. Domain Examples

The following examples illustrate how categorical and numerical data appear across three engineering and science domains, and why a value is meaningless without its variable name, unit, and experimental context.

Table 1 is referenced below and summarizes one categorical example, one numerical example, and the reason unit or context matters for each domain.

Table 1. Categorical and numerical data examples by domain.

Domain	Categorical Example	Numerical Example	Why Unit / Context Matters
Biochemistry	Sample Type: blood, saliva or urine	Glucose concentration (mg/dL), Testosterone concentration (ng/dL)	A value such as 95 is meaningless unless it is identified as the glucose concentration measured in mg/dL
Electronics	Semiconductor type: Silicon, Germanium, Gallium Arsenide	Forward Voltage Drop (V)	0.7 is meaningful only when it is identified as the forward voltage drop measured in volts (V), associated with a specific semiconductor material
Mechanical	Material type: Steel, Aluminium, Cast Iron	Tensile Strength (MPa)	450 is meaningful only when it is identified as tensile strength measured in megapascals (MPa), associated with a specific material

4. Pre-Analysis Checks

4.1 What to Inspect Before Analysis

Before analysing a dataset, the following aspects should be inspected:

- Verify that all variables have clear and descriptive names.
- Ensure that the units of measurement are consistent across the dataset.
- Check for missing, incomplete, or duplicate values.
- Identify any outliers or values that appear unrealistic.
- Confirm that categorical variables use consistent labels and formatting.
- Validate that numerical values fall within expected ranges.
- Check for invalid observations
- Check for duplicated observations
- Check for missing variables in observations

4.2 Questions to Ask a Data-Providing Team

Before analysing data received from another team, the following clarifying questions should be asked:

- What does each variable represent, and what are its units of measurement?
- How was the data collected?

- How diverse was the sample size?
- What do the categorical labels or codes represent?
- Are there any missing, estimated, or intentionally excluded values?
- Has the dataset already been cleaned, filtered, or transformed?
- Are there any known errors, limitations, or quality issues in the data?
- What is the purpose of the analysis, and what questions should it answer?
- Are there any assumptions or experimental conditions that should be considered during analysis?

5. Domain Thinking Reflection

A dataset is more than just a collection of numbers because each value represents a specific measurement taken under conditions. A numerical value becomes meaningful only when it is associated with its variable name, unit of measurement, experimental conditions, and domain context. Without this information, the data can be easily misinterpreted, leading to incorrect analysis and conclusions.

For example, in electronics, a value of 5 is meaningless unless it is identified as 5 volts (V) measured across a circuit component under specified operating conditions. Therefore, understanding the context behind the data is just as important as analysing the values themselves.

6. Conclusion

Working through this task made it clear that most analysis mistakes happen before any code gets written. They come from misreading what a column represents, or ignoring the unit attached to a value. A pH reading and a voltage reading can both be "7," but treating them as comparable numbers would be wrong in any domain.

The Mars Climate Orbiter example makes the point better than any abstract explanation could: two teams working with correct numbers still produced a failure because those numbers were tied to different unit systems.

The salary example makes a similar point about summary statistics. A mean can be technically correct and still tell you almost nothing about a typical case.

Going into pandas, the checklist in Section 4 is what actually matters day to day: know what each column represents, confirm the units, and check for missing or inconsistent values before trusting any statistic computed from them.

References

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